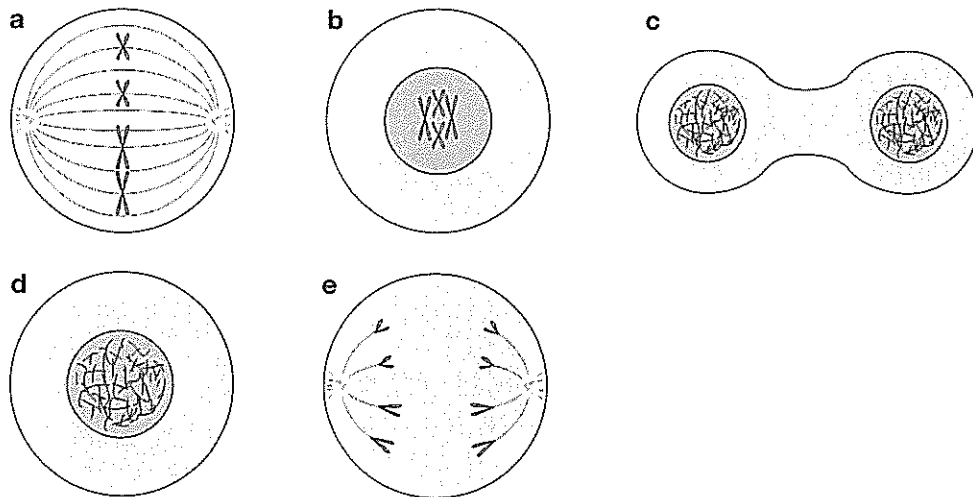


Science understanding



The diagram represents five stages of mitosis. However, they are not in the correct order.



The following captions A–E are descriptions of the five stages of mitosis shown. The descriptions are not in the correct order.

- A The chromatids separate. The spindle fibres contract, pulling the chromosomes to opposite poles of the cell.
 - B Separate chromosomes become visible. Each chromosome comprises two chromatids.
 - C In the period between cell divisions, the DNA replicates.
 - D The nuclear membrane re-forms, enclosing the chromosomes into a new nucleus at each pole. Division of the nucleus is now complete. The cytoplasm then divides, resulting in two identical daughter cells.
 - E The membrane surrounding the nucleus breaks down. The spindle appears, extending from the poles of the cell to each chromosome. The chromosomes line up across the equator of the cell.
- 1 (a) In the table below, redraw each stage of the diagram shown to **demonstrate** their correct order.
- (b) **Identify** the letter of the correct caption for each stage.

First stage	Second stage	Third stage	Fourth stage	Fifth stage
Caption:	Caption:	Caption:	Caption:	Caption:

Science understanding



Draw a line to **identify** the diagram of the stage of meiosis that matches each description.

1		a	The nuclear membrane breaks down and pairs of double-stranded chromosomes line up on the equator of the cell with spindle fibres attached.
2		b	The nuclear membranes form and the cytoplasm divides to produce four new cells. Each cell now contains the haploid number of chromosomes. These cells are the gametes.
3		c	In the period between cell divisions, the DNA replicates.
4		d	The spindle contracts, pulling the chromatids apart towards the poles of the cells.
5		e	At the end of the first part of the division, nuclear membranes may re-form. The chromosomes at each pole are enclosed into new nuclei.
6		f	The spindle contracts, drawing one chromosome from each pair to opposite poles of the cell. At this stage, each chromosome is still two chromatids.
7		g	The DNA becomes visible as separate chromosomes each of which comprises two chromatids.
8		h	After the first part of the division, a new spindle forms at right angles to the first. The new spindle attaches to the chromosomes that have lined up on the equator of the cell.